

## Pattaraphon Kenny Wongchamcharoen

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| EDUCATION | <b>University of California, Berkeley</b><br>B.S. Industrial Engineering & Operations Research<br>Minors in Data Science and Mathematics<br>GPA: 3.97/4.00 | Expected May 2027 |
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| RESEARCH<br>INTERESTS | <b>Methods:</b> Stochastic Modeling and Queueing Theory; Optimization; Game Theory; Statistical and Empirical Methods; Applied Machine Learning<br><br><b>Applications:</b> Human-AI Operations; AI Reliability in Operations; Service and Transportation Systems; Platform Operations and Marketplace Design |
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| PUBLICATIONS &<br>WORKING PAPERS | <ul style="list-style-type: none"> <li>[1] Do Large Language Models (LLMs) Understand Chronology?<br/> <b>Wongchamcharoen, P. K.</b>, &amp; Glasserman, P.<br/> <i>Proceedings of the 2026 AAAI Conference on Artificial Intelligence (AAAI-26)</i><br/>           Oral – Student Abstract &amp; Poster Program (Top 11%)<br/>           Poster – <i>AI4TS: AI for Time Series Analysis (AAAI-26 Workshop)</i><br/> <a href="#">[proceedings]</a> <a href="#">[arXiv]</a> <a href="#">[code]</a> </li> <li>[2] CentaurBench: Benchmarking LLM Capabilities on Augmenting vs. Automating Real-World Work Tasks<br/> <b>Wongchamcharoen, P. K.</b>, Gulati, K., Fong, M. M., &amp; Nagaraj, A.<br/> <i>NBER Working Paper #35663</i><br/>           Poster – <i>NeurIPS 2026 Workshop on Agentic AI Benchmarks and Applications for Enterprise Tasks (AABA4ET)</i><br/> <a href="#">[nber]</a> <a href="#">[arXiv]</a> <a href="#">[project dashboard]</a> </li> <li>[3] On-Off Systems with Strategic Customers<br/>           Sun, Y., <b>Wongchamcharoen, P. K.</b>, Liu, Z., &amp; Yan, C.<br/> <i>Revise and Resubmit (R&amp;R) at Operations Research</i><br/> <a href="#">[ssrn]</a> </li> <li>[4] Flex or Fast? Incentive-Compatible Demand Allocation in Destination-Mode Ride-Hailing Networks<br/> <i>In preparation</i> </li> </ul> |
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| OTHER RESEARCH<br>PROJECTS | <ul style="list-style-type: none"> <li>[1] Data-Driven Evaluation of Board of Directors Effectiveness<br/> <b>Wongchamcharoen, P. K.</b>, Stringer, C., Hwang, B., Liu, I., &amp; Li, K.<br/> <i>Best Data Visualization, Berkeley CDSS Data Discovery Symposium 2025</i><br/> <a href="#">[poster]</a> </li> <li>[2] Mixed-Integer Linear Program for Options Pricing and Portfolio Optimization<br/>           Bin Abdulla, Q. M., <b>Wongchamcharoen, P. K.</b>, Jamari, A., &amp; Lee, J.<br/> <i>1st Runner-up, Wells Fargo &amp; IEOR Bay Area Decision Sciences Summit 2025</i><br/> <a href="#">[poster]</a> <a href="#">[code]</a> </li> <li>[3] Calyber: A Ridesharing Game<br/>           Shen, Y., Yan, C., &amp; Yan, J.<br/> <i>INFORMS Transactions on Education</i>, 26(3):218–225, 2026<br/>           Research assistant contribution; Runner-up, 2025 INFORMS Case Competition<br/> <a href="#">[article]</a> <a href="#">[award]</a> </li> </ul> |
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## TALKS & PRESENTATIONS

Sun, Y., **Wongchamcharoen, P. K.**, Liu, Z., and Yan, C. (2026). “On-Off Systems with Strategic Customers.” Scheduled invited presentation at the *2026 INFORMS Annual Meeting, MSOM Invited Session* in San Francisco, CA.

**Wongchamcharoen, P. K.**, Gulati, K., Fong, M. M., and Nagaraj, A. (2026). “CentaurBench: Benchmarking LLM Capabilities on Augmenting vs. Automating Real-World Work Tasks.” Presentation at the *2026 Wharton Generative AI & Business Conference* in San Francisco, CA.

**Wongchamcharoen, P. K.** (2026). “AI Is Getting Smarter... Are We Cooked?: Research Perspectives on the Capabilities, Limits, and Future of AI.” Invited talk at the *Berkeley Engineering Summer Machine-Learning & AI Research Training (BeSMART) Program* in Berkeley, CA. [slides]

**Wongchamcharoen, P. K.** and Glasserman, P. (2026). “Do LLMs Understand Chronology?” Presentation at the *2026 IISE Annual Conference* in Arlington, TX.

**Wongchamcharoen, P. K.** and Glasserman, P. (2026). “The Temporal Blind Spots of AI: Implications for IEOR Applications.” Presentation at the *Berkeley IEOR Annual Advisory Board Meeting* in Berkeley, CA. [press]

**Wongchamcharoen, P. K.** and Glasserman, P. (2026). “Do LLMs Understand Chronology?” Oral presentation at the *40th Annual AAAI Conference on Artificial Intelligence* in Singapore. [poster] [talk]

## RESEARCH EXPERIENCE

**UC Berkeley Industrial Engineering & Operations Research** Berkeley, CA  
*Research Assistant*

*Advisor: Prof. Chiwei Yan*

Jan 2024 – Present

- Extended the on-off dynamic queueing model with strategic customers to multi-server systems; reformulated the nonconvex throughput problem as a linear program and proved theoretical results on throughput scaling, demand saturation, and optimal service allocation under exogenous on-off policies, including a closed-form linear-scaling threshold via an explicit dual certificate; contributed to the revision of *On-Off Systems with Strategic Customers* (R&R, *Operations Research*).
- Designed and validated a discrete-event simulator with Poisson arrivals, strategic customer joining, exhaustive service, and counterfactual wait tracking; developed learning dynamics and algorithms for endogenous equilibrium computation; evaluated stochastic-to-fluid convergence and out-of-sample policy performance.
- Estimated rider patience and recovered suppressed demand using a hierarchical Bayesian mixed-effects model with Bay Area Rapid Transit (BART) Yellow Line origin–destination and GTFS data; optimized reusable dwell-time policies across 48 station-direction queues using sample-average approximation, projecting peak-period throughput gains of up to 24%; invited to present in an MSOM session at the 2026 INFORMS Annual Meeting and contributed to securing a \$40,729 UC ITS research grant for passenger-centric BART schedule optimization.
- Deployed *Calyber*, a 2025 INFORMS Case Competition runner-up, in a graduate supply-chain course; developed Python notebooks and teaching materials; guided teams in designing pricing and matching policies using a dataset of more than 10,000 Chicago rider arrivals, supporting the case’s subsequent publication in *INFORMS Transactions on Education*; built an AI-agent competitor to benchmark agentic policies against student teams on profit, conversion, and matching.

**UC Berkeley Haas School of Business**

Berkeley, CA

*Research Assistant, Data Innovation and AI Lab (DIAL)**Advisor: Prof. Abhishek Nagaraj*

Jan 2024 – Present

- Led a four-person research team to conceptualize and develop *CentaurBench*, a benchmark distinguishing an LLM’s ability to automate professional work from its ability to augment another agent.
- Designed and ran experiments across seven professional tasks and ten frontier models, finding that automation and augmentation rankings diverge; NBER Working Paper; presented at the 2026 Wharton Generative AI & Business Conference.
- Engineered a multi-provider asynchronous Python evaluation pipeline across six model providers; built a reproducible LLM-as-judge harness using anonymized round-robin comparisons, randomized response ordering, micro-rubrics, four-model judge panels, and *leave-family-out* masking; validated 15,000+ scored comparisons using rank correlations, inter-judge agreement, Krippendorff’s alpha, and sensitivity analysis; created a public reproducibility dashboard and repository auditing outputs, rankings, rationales, and replication diagnostics.
- Supported research on generative AI and online platforms, building logistic regression and random forest classifiers to distinguish human from AI-generated OpenStreetMap contributions.

**Columbia Business School, Decisions, Risk, and Operations** New York, NY*Summer Research Fellow**Advisor: Prof. Paul Glasserman*

Jun 2025 – Aug 2025

- Selected as 1 of 15 interns from more than 1,000 applicants for Columbia Business School’s Summer Research Internship Program to investigate whether large language models (LLMs) understand chronology.
- Built and deployed a reproducible Python pipeline using GPT-4.1, o3, Claude 3.7 Sonnet, and other frontier LLM APIs to simulate event-ordering tasks and evaluate chronological reasoning at scale across increasingly complex task families: basic sorting, conditional sorting, and anachronism detection; developed a statistical evaluation framework comparing reasoning models using Spearman rank correlation, Kendall rank correlation, and complementary rank-based diagnostics across model families and task configurations.
- Published the findings in the AAAI-26 proceedings and presented them orally in the Student Abstract Program (top 11%) and as a poster at the AI4TS: AI for Time Series Analysis Workshop; invited to present at the 2026 IISE Annual Conference and Berkeley IEOR Annual Advisory Board Meeting; cited in OpenAI’s 2026 paper, “Scaling social science research” [*working paper*].

**UC Berkeley School of Law**

Berkeley, CA

*Student Researcher**Advisor: Prof. Frank Partnoy*

Jan 2025 – May 2025

- Scraped SEC DEF 14A board skills matrices; applied t-SNE clustering and predictive modeling to construct data-driven measures of board effectiveness.

**UC Berkeley Haas School of Business**

Berkeley, CA

*Research Assistant**Advisor: Prof. David Evan Harris*

Aug 2023 – Jan 2024

- Conducted literature review on AI ethics and governance.

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| PROFESSIONAL<br>EXPERIENCE  | <b>Beats by Dre</b> , Data Analytics Extern, Culver City, CA Jun–Aug 2024<br><b>Xcube Capital</b> , Product Development & Analytics Intern, Singapore May–Aug 2024<br><b>MENTOR Global Consultants</b> , Technical Product Mgmt. Intern Jan–Apr 2024<br><b>GUESS, Inc.</b> , Strategy Contract Consultant Aug–Dec 2023                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| HONORS AND<br>AWARDS        | Tau Beta Pi Scholarship (Tau Beta Pi Association; \$2,000) 2026<br>Bonder Foundation Scholarship (INFORMS; \$400) 2026<br>Dean’s Lists (UC Berkeley College of Engineering) 2023–2026<br>CBS Research Internship Grant (Columbia Business School; \$7,500) 2025<br>Best Poster: Data Visualization Award (UC Berkeley CDSS) 2025<br>1st Runner-up, Bay Area Decision Sciences Summit (Wells Fargo; \$600) 2025<br>Cal Alumni Leadership Award Finalist (Cal Alumni Association) 2024<br>College of Engineering Mentorship Grant (UC Berkeley; \$850) 2024<br>ASEAN Scholarship (Singapore Ministry of Education; full 4-year award) 2018                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| TEACHING<br>EXPERIENCE      | <b>Tutor &amp; Grader</b> , University of California, Berkeley Fall 2024–Present <ul style="list-style-type: none"> <li>• IEOR 145: Fundamentals of Revenue Management (Prof. Rajan Udwani)</li> <li>• IEOR 142A: Machine Learning &amp; Data Analytics I (Prof. Phillip Kerger)</li> <li>• IEOR 172: Probability &amp; Risk Analysis for Engineers; IEOR 120: Principles of Engineering Economics (Prof. Daniel Pirutinsky)</li> <li>• CS 88: Computational Structures in Data Science (Prof. John DeNero)</li> <li>• Tutor for lower- and upper-division courses in mathematics, data science, computer science, and IEOR, Berkeley’s Center for Access to Engineering Excellence (CAEE)</li> </ul><br><b>Primary Lecturer</b> , Berkeley Engineering GLOBE Center Fall 2025–Present <ul style="list-style-type: none"> <li>• Developed and taught <i>Foundations of Machine Learning</i>, an introductory machine-learning course for underserved Bay Area high-school students (Program Advisor: Prof. Phillip Kerger). [<i>course website</i>]</li> </ul><br><b>Teaching Assistant</b> , University of California, Berkeley Spring–Summer 2026 <ul style="list-style-type: none"> <li>• IEOR C253: Supply Chain &amp; Logistics Management (Prof. Chiwei Yan)</li> <li>• Berkeley Summer Machine Learning &amp; AI Research Training Program (BeSMART); mentored student machine learning research projects and presentations, created assignments, and guest lectured on coding and agentic AI (Prof. Phillip Kerger)</li> </ul> |
| SERVICE AND<br>LEADERSHIP   | <b>Reviewer</b> , NeurIPS 2026 Workshop AABA4ET, 2026<br><b>Institute of Industrial &amp; Systems Engineers at Berkeley</b><br>Senior Advisor Fall 2025 – Present<br>Director of Programs and Events Fall 2023 – Spring 2024<br><b>Singapore Students Association at Berkeley</b><br>Vice President of External Relations Spring 2024 – Fall 2024                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| PROFESSIONAL<br>MEMBERSHIPS | Institute for Operations Research and the Management Sciences (INFORMS); Institute of Industrial & Systems Engineers (IISE); Tau Beta Pi Engineering Honor Society                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| SKILLS &<br>LANGUAGES       | Python (NumPy, Pandas, Matplotlib, scikit-learn, Pyomo, TensorFlow); Java; SQL; Apache Spark; LLM APIs (OpenAI, Anthropic, Google Gemini); HuggingFace; L <sup>A</sup> T <sub>E</sub> X<br>English (Fluent); Thai (Native); Mandarin Chinese (Elementary)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |